

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I R.HARI KRISHNAN(192565075) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:R.Hari krishnan

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

COMPUTER NETWORKS

ASSESSMENT TOOL 2

DETAILED MODEL ANSWERS

Question 1 – Smart Campus Subnet Allocation

Scenario:

The Smart Campus has been allocated the network 192.168.100.0/24. Different departments require different subnet sizes. Variable Length Subnet Masking (VLSM) is used to allocate only the required number of addresses to each department, reducing IP address wastage and improving network efficiency.

Subnet Calculations:

School of Computing (/25)

Subnet Mask: 255.255.255.128

Network Address: 192.168.100.0

Broadcast Address: 192.168.100.127

Valid Hosts: 192.168.100.1 – 192.168.100.126

Usable Hosts: 126

School of Electronics (/26)

Subnet Mask: 255.255.255.192

Network: 192.168.100.128

Broadcast: 192.168.100.191

Valid Hosts: 192.168.100.129 – 192.168.100.190

Usable Hosts: 62

School of Mechanical Engineering (/27)

Subnet Mask: 255.255.255.224

Network: 192.168.100.192

Broadcast: 192.168.100.223

Valid Hosts: 192.168.100.193 – 192.168.100.222

Usable Hosts: 30

Unused Address Range:

192.168.100.224 – 192.168.100.255 (32 addresses reserved for future expansion).

Engineering Interpretation:

The Computing department receives the largest subnet because it has the highest number of devices.

Electronics receives a medium-sized subnet, while Mechanical Engineering receives a smaller subnet.

VLSM improves scalability, minimizes address wastage and allows future departments to be added without redesigning the network.

Question 2 – Company Network Design using VLSM

The company has received the private block 10.10.0.0/16. The administrator allocates different subnet sizes according to department requirements.

Software Development (/24)

Network: 10.10.0.0

Broadcast: 10.10.0.255

Host Range: 10.10.0.1 – 10.10.0.254
Usable Hosts: 254

Quality Assurance (/25)
Network: 10.10.1.0
Broadcast: 10.10.1.127
Host Range: 10.10.1.1 – 10.10.1.126
Usable Hosts: 126

Human Resources (/26)
Network: 10.10.1.128
Broadcast: 10.10.1.191
Host Range: 10.10.1.129 – 10.10.1.190
Usable Hosts: 62

Executive Management (/27)
Network: 10.10.1.192
Broadcast: 10.10.1.223
Host Range: 10.10.1.193 – 10.10.1.222
Usable Hosts: 30

Remaining Unused Range:
10.10.1.224 onward remains available within the allocated /16 network.

Conclusion:

Using VLSM allows the organisation to allocate addresses based on actual requirements instead of wasting hundreds of addresses. The remaining address space can later be assigned to new departments without affecting the existing network.

Question 3 – IPv6 Migration

Original IPv6 Address:
2001:0db8:abcd:0000:0000:0000:0000/64

Compressed Form:
2001:db8:abcd::/64

Expanded Form:
2001:0db8:abcd:0000:0000:0000:0000:0000

Network Prefix:
/64 means the first 64 bits represent the network and the remaining 64 bits represent the interface identifier.

Total Host Addresses:
 $2^{64} = 18,446,744,073,709,551,616$ possible interface addresses.

Advantages of IPv6:

- Extremely large address space.
- Supports automatic address configuration.
- Better routing efficiency.
- Improved security through IPSec support.
- Ideal for hospitals with thousands of IoT sensors and medical devices.

Question 4 – Routing Decision

Routing is the process of forwarding packets based on the destination IP address. Routers compare the destination address with entries in the routing table using the longest-prefix-match rule.

Routing Table Entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination Address:

192.168.2.45

Analysis:

The destination belongs to the network 192.168.2.0/24 because the first 24 bits match that subnet.

Matching Entry:

192.168.2.0/24

Forwarding Decision:

The packet is sent through the interface or next-hop configured for the 192.168.2.0/24 network.

Default Route:

If none of the routing entries match, the router forwards the packet through the default route (0.0.0.0/0), which usually points to the ISP or upstream router.

Importance:

Routing tables reduce unnecessary broadcasts and ensure packets reach the correct destination with minimum delay.

Question 5 – Layer-3 Router LAN Configuration

Administration LAN

Network Address: 192.168.10.0/26

Broadcast Address: 192.168.10.63

Host Range: 192.168.10.1 – 192.168.10.62

Gateway: 192.168.10.1

Example Hosts:

PC1 – 192.168.10.2

PC2 – 192.168.10.3

PC3 – 192.168.10.4

Finance LAN

Network Address: 192.168.10.64/26

Broadcast Address: 192.168.10.127

Host Range: 192.168.10.65 – 192.168.10.126

Gateway: 192.168.10.65

Example Hosts:

PC1 – 192.168.10.66

PC2 – 192.168.10.67

PC3 – 192.168.10.68

Working:

Each host uses its configured default gateway to communicate with devices outside its own LAN. The Layer-3 router receives packets from one subnet, examines the destination network, consults its routing table and forwards the packet to the correct destination subnet.

Real-World Application:

This design is commonly used in companies, universities and hospitals where different departments are separated into different subnets for security, performance and easier management.

Overall Conclusion:

The scenario demonstrates practical applications of subnetting, VLSM, IPv6 addressing and routing. Proper IP planning reduces address wastage, improves security, simplifies troubleshooting and ensures efficient communication between different departments in modern computer networks.